

Project Executable Files

Date	2 JULY 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	N/A
5	Environment configuration file (.env.example or similar)	N/A
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	N/A
8	Dockerfile / Containerization config (if applicable)	N/A
9	Test files and test results	Yes

10	Demo video or walkthrough (if required)	Yes
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Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

OPTICROP/
— data/
— Crop_recommendation.csv
— models/
— crop_model.joblib
— scaler.joblib
— templates/
— index.html
— static/
— style.css
— script.js
— app.py
— train.py
— notebook.ipynb
— requirements.txt
— README.md
— Project Documentation.pdf

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Deployed (Runs on Localhost)
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	[e.g., Render, Vercel, AWS, Heroku, Azure]

Repository Link	
Demo Video Link	

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

1. Install Python 3.9 or later. 2. Open the project in Visual Studio Code. 3. Install dependencies using: <pre>pip install -r requirements.txt</pre> 4. Train the Machine Learning model: <pre>python train.py</pre> 5. Start the Flask application: <pre>python app.py</pre> 6. Open your browser and visit: <pre>http://127.0.0.1:5000</pre> 7. Enter soil nutrient and weather values. 8. Click Predict to receive the recommended crop.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Works only on Localhost	Can be deployed to Render or Railway in the future
2	No user authentication	Can be added in future versions
3	Uses static CSV dataset	Can be connected to a real-time agricultural database